

Shyamal Das

PhD Scholar, Electrical Engineering (Power Systems)
IIT Bhubaneswar, Odisha, India

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Google Scholar — ORCID — ResearchGate

Research Identity

Protection and control of inverter-dominated microgrids; grid-forming and grid-following inverter dynamics; islanding detection; PHIL/HIL based validation; real-time simulation using RTDS and Typhoon HIL; coordinated protection schemes for DER-rich systems.

Education

Ph.D. in Electrical Engineering (Power and Energy Systems)

Indian Institute of Technology Bhubaneswar

2022–Present

Thesis: Development of Coordinated Protection and Control Scheme for Microgrid

M.Tech in Electrical Engineering (Power and Energy Systems)

IEST Shibpur

2020–2022

CGPA: 9.54/10

Thesis: Islanding Detection in Microgrid Using Decision Tree Classifier

B.Tech in Electrical Engineering

Ramkrishna Mahato Government Engineering College, MAKAUT

2016–2020

DGPA: 8.69/10

Thesis: Load Data Corruption Identification and Load Forecasting using ANN and DCT-LSTM

Publications

Journal

1. **S. Das**, P. D. Achlerkar and S. R. Samantaray, “Hybrid Islanding Detection Based on Voltage THD and Non-Power Frequency Impedance in Inverter-Interfaced Microgrid,” in *IEEE Transactions on Smart Grid*, Early Access, 2026, doi: 10.1109/TSG.2026.3656067.

Conferences

1. **S. Das**, P. D. Achlerkar, S. R. Samantaray, “A Remote Current Injection Based Islanding Detection Technique for Microgrid,” *2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC)*, Goa, India, 2025.
2. **S. Das**, P. D. Achlerkar, S. R. Samantaray, “A Harmonic Injection Based Fault Detection Scheme for Inverter Interfaced Islanded Microgrid,” *2024 23rd National Power Systems Conference (NPSC)*, Indore, India, 2024, pp. 1-6, doi: 10.1109/NPSC61626.2024.10987085.
3. **S. Das**, S. Chowdhury, J. K. Chandra and S. Sen, “Short Term Missing Data Prediction of Power System Using Discrete Cosine Transform and Simple Continued Fraction Compressed Long Short Term Memory Network,” *2022 IEEE Calcutta Conference (CALCON)*, Kolkata, India, 2022, pp. 109-112, doi: 10.1109/CALCON56258.2022.10059693.
4. **S. Das**, A. De, “Islanding Detection in Microgrid Using Decision Tree Pattern Classifier,” in *Computational Intelligence in pattern recognition, 2022, lecture notes in networks and systems*, vol. 480. Springer. doi: 10.1007/978-981-19-3089-8_12.
5. **S. Das**, S. Kumar, D. Saha and M. Kumar, “Ensuring Stable System Performance with HHO Optimized Controller in a Deregulated AGC System Incorporating EVs and Wind Generator,” *2021 2nd International Conference on Communication, Computing and Industry 4.0 (C2I4)*, Bangalore, India, 2021, pp. 1-6, doi: 10.1109/C2I454156.2021.9689277.

Sponsored Research Experience

SERB Sponsored Research Project — Project Scholar

First 2 years of PhD

Worked on microgrid protection and control under funded project framework; contributed to modeling, validation, and deliverable reporting.

Experimental & Lab Infrastructure Experience

- Power Hardware-in-the-Loop (PHIL) testing
- RTDS real-time simulation platform
- Typhoon HIL setup and controller testing
- SPS amplifier interfacing
- Real inverter (5 kW) experimental validation
- Microgrid protection testing workflows

Teaching & Training Experience

- Delivered lecture for OPTCL trainee engineers — Load Flow Studies
- Invited lecture (planned) — Microgrid / Protection topic
- Teaching assistance in power systems labs and courses

Academic Service

- Reviewer — IEEE Transactions on Smart Grid
- Reviewer — IEEE PETEA Conference

Technical Skills

Simulation & EMT: MATLAB/Simulink, PSCAD, DIgSILENT

Real-Time / HIL: RTDS, Typhoon HIL

Programming: MATLAB, Python, C, Java

Domains: Microgrid protection, inverter control, islanding detection

Training

- South Eastern Railway, Adra Division — Vocational Training (2018)
- Mejia Thermal Power Station (DVC) — Vocational Training (2019)

Certifications

- Machine Learning and Applications — IEST Shibpur
- Microgrids and Electric Vehicles — NWSGT Workshop

Achievements

- GATE EE 2020 — Score 405
- Best Student Award — School level

Research Interests

Microgrid protection, inverter-dominated grids, grid-forming control, PHIL validation, DER protection coordination, AI in power systems.

Personal Details

Date of Birth: 1 July 1997

Languages: English, Hindi, Bengali

Hobbies: Reading, poetry writing, chess, gardening

References

Available upon request (or list 3 referees here).